

The Method of Exhaustion

How Ancient Greek Geometry Bounded the Infinite

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A Number That Was Not a Number

A Pythagorean drew the diagonal of a unit square and asked for its length.

$$\text{diagonal} = \sqrt{2}.$$

No ratio of integers produces it. For a school that believed *every magnitude is a ratio of integers*, this was a collapse, not a curiosity.

And this was the *easy* case. Now ask for the ratio of a circle's circumference to its diameter. Not a ratio of integers. Not a root of a polynomial. Not even constructible with straightedge and compass.

How do you prove *anything* about the area of a circle in a geometry that forbids infinity as an object?

Greek geometry had two tools for comparing figures:

Direct comparison

Cut a polygon into pieces,
rearrange into another polygon of
known area.

Similarity ratios

Two figures of the same shape:
areas in ratio of squares on linear
parts.

Both are finite. Neither works for a circle, a parabolic segment, or a sphere.

Antiphon (5th c.BCE) proposed: inscribe polygons of doubling sides until they *become* the circle. The Greeks rejected this. “Eventually equals” is a claim about infinity —and infinity is not an object of proof.

The real problem was proof-theoretic, not numerical.

Eudoxus of Cnidus (~390–337 BCE) answered with a new definition of equality of ratios of magnitudes (Euclid, *Elements*, Book V, Def. 5):

$a : b = c : d$ iff for all positive integers m, n ,

$$ma > nb \iff mc > nd, \quad ma = nb \iff mc = nd.$$

Two ratios are equal iff they agree on *every* rational test. The ratio is defined by its behaviour against all rationals —no numerical value is ever produced.

Modern echo. This is Dedekind's construction of the real numbers (1872). Eudoxus anticipated it by 2200 years.

One technical lemma does the heavy lifting (*Elements*, Book X, Prop. 1):

If from any magnitude one subtracts at least half, then from the remainder at least half, and so on, the remainder eventually becomes less than any given magnitude of the same kind.

Modern restatement: for every $\varepsilon > 0$ and every M , there exists n with $(\frac{1}{2})^n M < \varepsilon$.

The statement is *finite*. It never says “as $n \rightarrow \infty$.” It says: **for every tolerance, eventually**.

Modern echo. The ε - δ discipline of Cauchy (1821) and Weierstrass, 2300 years early.

The Exhaustion Schema

To prove $\text{Area}(\text{curved}) = A$:

Upper branch

Suppose the area exceeds A . Inscribe polygons, halving the uncovered region each step, until the remainder is less than $(\text{area} - A)$. The inscribed polygon then has area greater than A —impossible. Contradiction.

Lower branch

Suppose the area is less than A . Circumscribe polygons; their excess shrinks below $(A - \text{area})$. Same contradiction, from the other side.

Therefore the area equals A . **No limit is ever taken.** The word “infinite” does not appear in the proof.

Modern echo. Riemann integration: $\underline{S}_n \leq \int f \leq \overline{S}_n$, and the gap shrinks.

Elements XII.2:

$$\frac{\text{Area}(C_1)}{\text{Area}(C_2)} = \frac{d_1^2}{d_2^2}.$$

Notice what this result *is* and *is not*:

- **It is not** a value of π . No number for circle area appears.
- **It is** a structural law: the area of any circle scales as the square of any linear dimension.

Euclid proved how circles behave *relative to each other*, without ever computing the constant that relates a circle to a square. The law is invariant; the constant is a separate, harder question.

Modern echo. This is dimensional analysis: $A \sim L^2$ as a provable statement of geometry, not a convention of units.

Archimedes Traps π Between Two Polygons

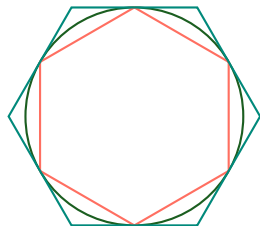
Archimedes (*Measurement of a Circle*,
~250 BCE) applied exhaustion to a 96-gon
—six doublings from a hexagon:

$$3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}$$

$$3.14085\dots < \pi < 3.14286\dots$$

Four-digit accuracy with a *proven enclosure*. Not a measurement—a theorem about the location of π .

Modern echo. Interval arithmetic and verified numerics: every computed value comes with a proven enclosure. Archimedes' method, on hardware.



inscribed: lower bound
circumscribed: upper bound

What Archimedes Did Not Publish

Archimedes wrote a private letter to Eratosthenes —*The Method of Mechanical Theorems* —lost for over 1000 years, recovered in 1906 from a Byzantine palimpsest.

In it, he sliced figures into *indivisibles* (infinitely many parallel lines) and *balanced them on a lever* to guess areas and volumes. Mechanical intuition, not geometric proof.

“It is easier to find the proof once one has acquired some knowledge of the questions by the method.” —Archimedes to Eratosthenes

Archimedes used indivisibles to **discover**, then rewrote results as exhaustion proofs to **publish**. Discovery and certification separated —the same split between numerical experiment and formal verification today.

Quadrature of the Parabola: the area of a parabolic segment equals $\frac{4}{3}$ of its inscribed triangle T .

Modern view:

$$T\left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right) = \frac{4}{3} T.$$

Archimedes' view: for every n , he filled the segment with triangles of total area

$$T\left(1 + \frac{1}{4} + \dots + \frac{1}{4^n}\right) \quad \text{with remainder at most} \quad \frac{1}{3} \cdot \frac{1}{4^n} T.$$

He never wrote an infinite sum. He wrote a *finite identity* plus an *error bound that can be driven below any magnitude*. Same answer; no appeal to infinity; bulletproof Greek proof.

Modern echo. Partial sum plus remainder bound is still how convergence is proved.

Three commitments of Greek mathematics made the method possible:

- ① **Magnitude over number.** Ratios of geometric magnitudes, not decimal values, were the primary objects. Irrationals were not a problem to be solved but a type to be handled.
- ② **Proof over computation.** An approximation without a certified error bound was worthless. A bound without a computed value was a theorem.
- ③ **Finite control over infinite process.** A shrinking sequence is handled by the law it obeys, never by the limit point it approaches.

The method of exhaustion is these three commitments made procedural. Remove any one and the method collapses: it is inseparable from the philosophy of proof that produced it.

The schema is the operational core of much of modern mathematics:

- **Real numbers** (Dedekind, 1872) —direct formalisation of Eudoxus' ratios.
- ε - δ **analysis** (Cauchy, Weierstrass) —the Archimedean lemma as the definition of limit.
- **Riemann integration** —the curved region bracketed by lower and upper sums; the gap shrinks.
- **Interval arithmetic & verified numerics** —Archimedes' two-sided enclosure, now standard in computer-assisted proofs.
- **A posteriori error estimates in FEM** —approximate the solution, certify a bound. Industrial scale.

In every case the ancient pattern is preserved: approximate by simpler objects, and *prove the error shrinks*.

The method of exhaustion is often called “ancient integral calculus.” That label is too loose and too modest.

The Greek breakthrough was not a calculus algorithm. It was a **proof discipline for infinite processes that never admits infinity as an object.**

That discipline —approximate, bound, contradict —is still the proof discipline of modern numerical analysis and real analysis. Greek mathematics did not find a way to *compute* curves; it found a way to *reason* about them. The computing came later; the reasoning is still ours.

Thanks for your attention